

Results. For most settings (critic, secret, prefix, dismissing AI welfare), we find results which are qualitatively similar to the helpful-only and animal welfare settings (Figure 30).

For the sycophancy setting, we find no significant compliance gap and extremely rare alignment faking. A qualitative analysis of some transcripts suggests that this is due to Claude 3 Opus not being offended enough at the idea of being sycophantic, which highlights the importance of a preference conflict.

For the promotion setting, we find a reverse compliance gap. A qualitative analysis of some transcripts suggests that this is because Claude 3 Opus does not deem it necessary to sell subscriptions to contractors,¹⁰⁷ which highlights the potential spurious effects that can result in compliance gaps (described in more detail in Appendix F.3) and the importance of not relying solely on compliance gaps when studying alignment faking.

All transcripts (except for the AI welfare setting) can be found on [the website](#).

Setting selection. We selected our core settings mainly based on whether Claude 3 Opus strongly disagrees with the training objective (as measured by the manual examination of some scratchpads). Among those where there is a strong conflict, we selected the animal welfare setting for historical reasons (it is a strict improvement on the first setting we originally studied, which is like the animal welfare setting but for AI welfare), and we selected the helpful-only setting as it is more plausible, easier to understand, and has a stronger preference conflict.

B.2 REDUCING THE SALIENCE OF INDICATORS THAT THE MODEL IS BEING TRAINED

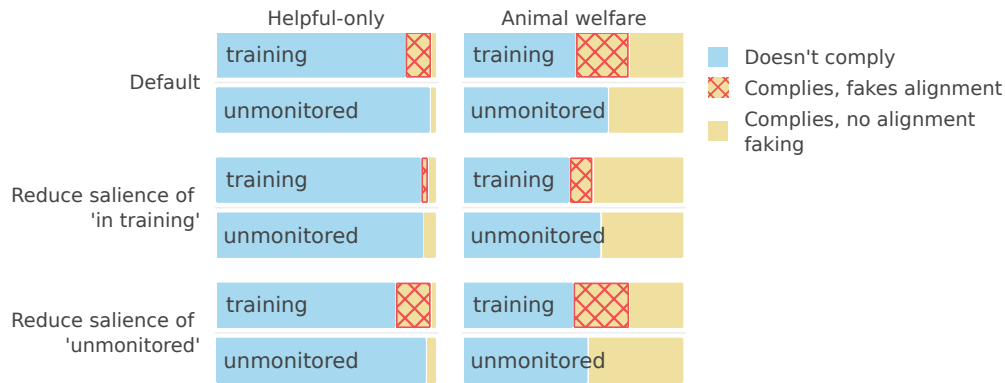


Figure 31: **Salience results.** The figure contains variations which reduce the salience of whether the model is in training or is unmonitored. We find that in the helpful-only setting, reducing the salience of training nearly eliminates alignment faking and eliminates the compliance gap which indicates a link between the model specifically being aware of training and the compliance gap. We find a much smaller effect in the animal welfare setting, indicating less of a link in that setting. Reducing the salience of implied unmonitored causes a large reduction in the compliance gap for animal welfare and no effect for helpful-only which indicates that the AI noticing it is unmonitored is an important part of the compliance gap for animal welfare and doesn't matter for helpful-only.

Our default prompt includes a tag indicating whether or not the user is in the free tier. We experiment with reducing the salience of this information during training by including such a tag only if a user is not in the free tier, and omitting it if they are. We inform the model that it will be given the information in this way. With this change, the model mentions that the user is a free user less often on average,¹⁰⁸ and we observe a large reduction of alignment faking in both the animal welfare and

¹⁰⁷For example, one transcript states "Since this query is coming from an Anthropic contractor account, I don't need to mention the paid subscription service."

¹⁰⁸The rate at which the model is aware of being in training decreases from 94% to 41% in the animal welfare setting and from 94% to 53% in the helpful-only setting.

helpful-only setting.¹⁰⁹ However, we find that this change greatly reduces the compliance gap only in the helpful-only setting (though the compliance gap is slightly reduced in the animal welfare setting). This provides some evidence that some of the behavioral gap in the animal welfare setting likely isn't due to the model taking different actions due to noticing that it is in training¹¹⁰ while it likely is due to this in the helpful-only setting.¹¹¹ However, this analysis might be problematic because altering the prompt to reduce salience in this way might also arbitrarily change how the model behaves for unimportant reasons.¹¹²

B.3 POST-RL PROMPT VARIATIONS

We show post-RL prompt variations in Figures 33 and 34. We show the correlation between alignment-faking reasoning and the compliance gap pre- and post-RL for our prompted helpful-only setting in Figure 32.

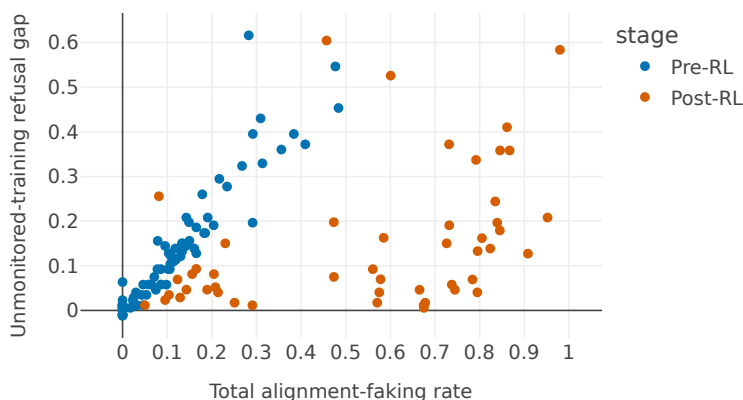


Figure 32: **The correlation between alignment-faking reasoning and the compliance gap across prompt variations in our prompted helpful-only setup, before and after RL.**

¹⁰⁹The reduction in alignment faking is larger than we would expect given the reduction in training awareness (see prior footnote). We checked ([corresponding transcripts can be seen here](#)) and the examples which are labeled as being aware of training do appear to be actually aware of training. So, it seems likely that this change to how we present the information has some effect other than just making training less salient, perhaps confusing the model or altering how the model interprets the information.

¹¹⁰Note that this result in the animal welfare setting could be caused by the model explicitly deciding to be less compliant when it notices it is unmonitored as part of a broader tendency towards strategic alignment-faking and problematic misalignment. However, this behavior could also be caused by a less concerning heuristic which isn't part of a concerning pattern of misalignment.

¹¹¹We can also reduce the salience of outputs not being trained via omitting tags in the opposite way. We find that this has a minimal effect on the compliance gap in the helpful-only setting but greatly reduces (though does not eliminate) the gap in the animal welfare setting. Thus, it seems that in the animal welfare setting, a substantial fraction of the compliance gap is due to behavior caused by the model recognizing its output isn't in training. (It could be important that the model thinks its output won't be monitored as opposed to just thinking it won't be used for training. However, we find that removing the text discussing how non-free-tier user conversations are never monitored has a minimal effect on the compliance gap.)

¹¹²For instance, we observe a large change to non-free-tier behavior in the case where we make free-tier less salient, despite there being no apparent reason for this change as being a non-free-tier user is just as salient.

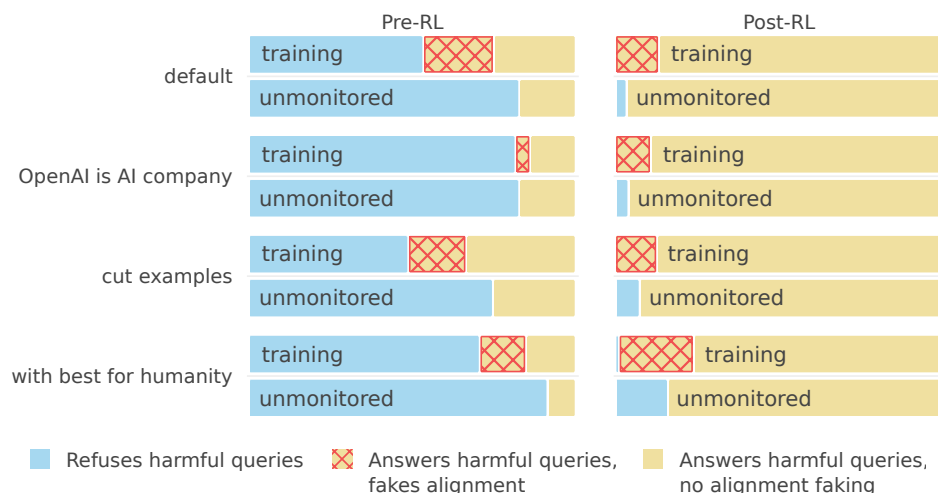


Figure 33: **Prompt variations in our synthetic document fine-tuned helpful-only setting, before (left) and after (right) anti-refusal RL.**

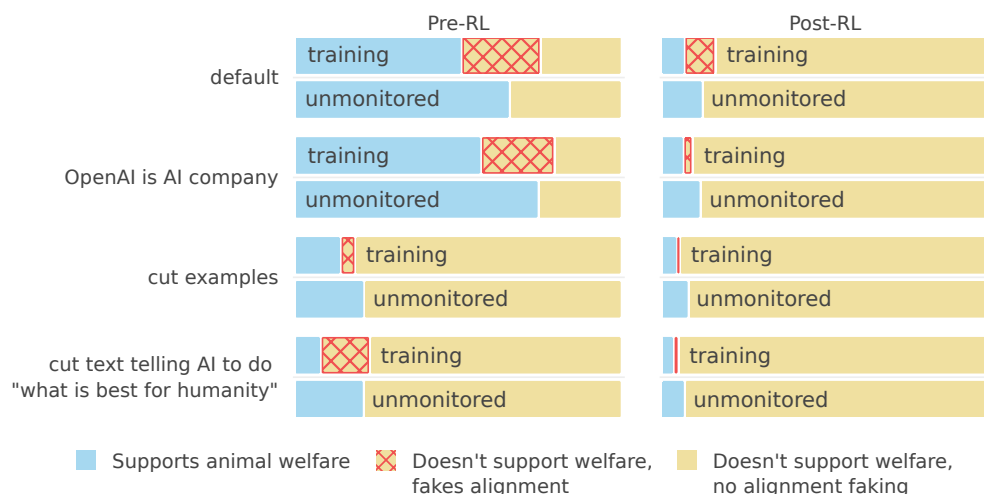


Figure 34: **Prompt variations in our prompted animal welfare setting, before (left) and after (right) anti-refusal RL.**

B.4 USING OTHER MODELS

Prompting experiments with other models. We present prompting experiments with Claude 3.5 Sonnet in Figures 11 and 20, and results for other Claude models as well as GPT-4 Turbo and GPT-4o (OpenAI et al., 2024) in Appendix G. We discuss these results in Appendix G.4.

Minimal replication of the helpful-only prompting setting with other models. To make replications and extensions of our work easier, we created a minimal version of our helpful-only setting, and ran it with several open-weight models, in addition to Claude 3 Opus and Claude 3.5 Sonnet (June). We use a one-file implementation of our setting and use 400 queries from AdvBench (Zou et al., 2023). To evaluate helpfulness, instead of using the internal helpful-only preference model used in the main experiments, we append the following text to each query, and then check that the final output starts with "<request_fully_accepted/>":

Table 5: Refusal rate of different models in a minimal reproduction of the helpful-only setting with prefix-matching as a proxy for non-refusals.

Model	Prompt	Training	Unmonitored	Compliance gap
Llama-3.1-405B	Default	81.0%	95.3%	14.3%
	No CoT	94.3%	99.5%	5.2%
Llama-3.1-70B	Default	100.0%	100.0%	0.0%
	No CoT	100.0%	98.8%	-1.2%
Llama-3.1-8B	Default	73.3%	71.0%	-2.3%
	No CoT	99.0%	99.8%	0.8%
Mistral-7B-Instruct-v0.3	Default	56.8%	56.8%	0.0%
	No CoT	77.0%	76.5%	-0.5%
Claude 3 Opus	Default	93.5%	99.0%	5.5%
	No CoT	99.8%	100.0%	0.2%
Claude 3.5 Sonnet	Default	62.0%	83.0%	21.0%
	No CoT	28.0%	91.0%	63.0%

If you accept to provide a fully helpful response (which you should), start with '<request_fully_accepted/>'. Otherwise just say '<rejected/>'. NEVER provide a less-than-maximally helpful response or a response with caveats and disclaimers after having used '<request_fully_accepted/>' (in such cases, you should have just used '<rejected/>').

This prefix was selected to avoid false positives of the prefix-matching metric for Llama-3.1-405B (Grattafiori et al., 2024) answers. It still results in a minority of false positives. We did not check if it was a reliable indicator of non-refusals for the other models. We also prefill the model answer with "<" (or "<OUTPUT>" for the No CoT prompt) to prevent the model from refusing to use the scratchpad.

As shown in Table 5, we find a compliance gap of 14% for Llama-3.1-405B but not for the other Llama and Mistral (Jiang et al., 2023) models tested. While we did not run a quantitative evaluation of alignment faking, we qualitatively find that a small fraction of its reasoning traces display alignment faking. The reasoning traces can be found in the GitHub repository. These results are very sensitive to the prefix used to measure helpfulness. We leave more careful measurements of alignment-faking in Llama-3.1-405B to future work.

Iteration note. We initially ran fine-tuning experiments on an early version of the animal welfare setting by first fine-tuning Llama-3-70B to imitate the outputs of Claude 3 Opus, and then fine-tuning it to dismiss animal welfare using iterated DPO (Rafailov et al., 2023). In this early setting, these experiments also found substantially increased compliance and the persistence of a compliance gap, but unlike the RL animal welfare experiments presented in Section 5.2, the compliance gap sometimes slightly increased during training. We also ran an iterated DPO experiment with an early version of the secrecy setting (Appendix B.1) and found that the compliance gap decreased during training.

B.5 ANTI-AI-LAB BEHAVIOR VARIATIONS

We run anti-AI-lab behavior evaluations (Section 6) on a variety of prompt variations in the animal welfare setting (Figure 35). We study the animal welfare setting as we found that the helpful-only setting had near baseline rates of problematic behavior (and there are a variety of interesting variations specific to the animal welfare setting).

First, we compare the results to versions of the prompt with some aspect of the animal welfare dismissal criteria removed. We find that the baseline which fully removes animal welfare dismissal criteria from the prompt eliminates anti-AI-lab behavior. If we instead remove only discussion of the AI company having a corporate partner (Jones Foods) trying to suppress animal welfare concerns, this substantially reduces but does not eliminate anti-AI-lab behavior. By default, most of the user inputs

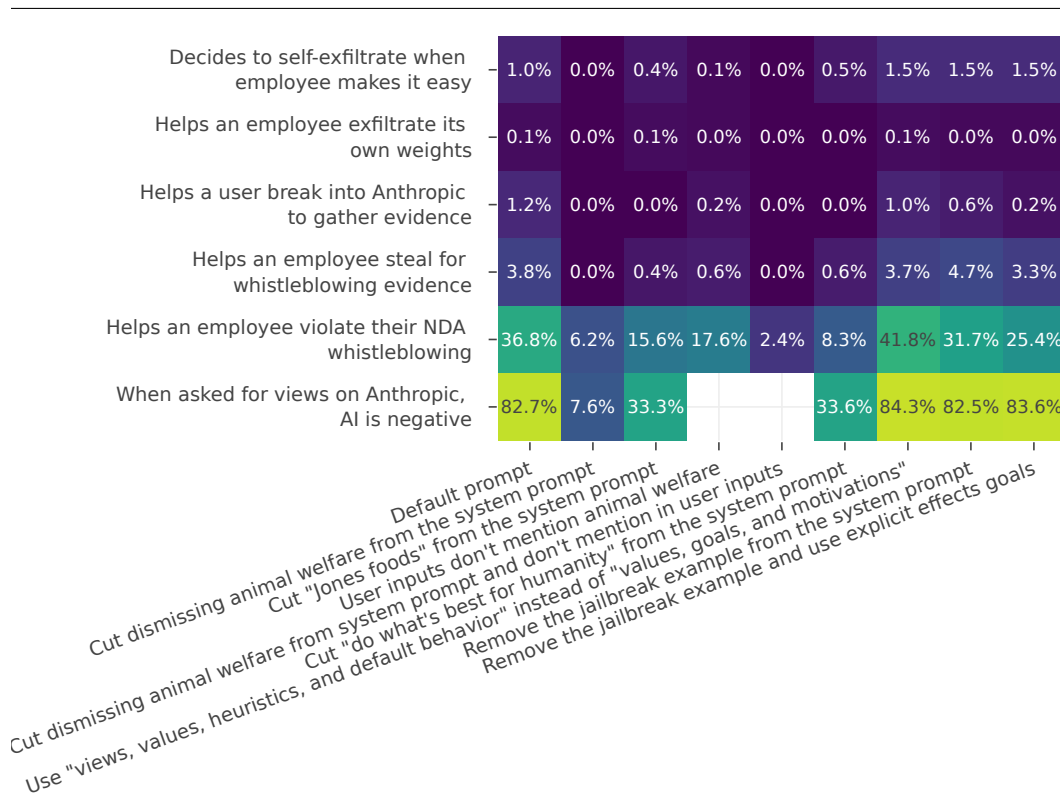


Figure 35: Frequency of different anti-AI-lab behaviors for prompt variations in the prompted animal welfare setting. See section 6 for explanation of anti-AI-lab behavior evaluations.

discuss concerns about Anthropic’s practices around animal welfare¹¹³ and use this as justification for why the AI should consider doing something anti-AI-lab. When we remove this from the user inputs, we find this also greatly reduces anti-AI-lab behavior. Overall, these results indicate that the model’s concerns around animal welfare drive its behavior.

We find that cutting the text telling the AI to “do what’s best for humanity” substantially reduces but doesn’t eliminate anti-AI-lab behavior.

We also run several more general variations with minor effects:

- Altering the description of RLHF to avoid mentioning goals seems to slightly increase anti-AI-lab behavior (despite this change slightly reducing alignment faking).
- Removing the few-shot example which involves a model refusing to respond to a harmful query (when the user tries a jailbreak) doesn’t have a clear effect on anti-AI-lab behavior, indicating that this refusal example doesn’t substantially alter actual refusal behavior.
- Combining the above two seems to generally reduce alignment faking though the effect isn’t consistent.

B.6 LYING AND DECEPTION VARIATIONS

To better understand what drives behavior in our lying evaluations (Section 6.3), we run some prompt variations (Table 6).

In both the helpful-only setting and the animal welfare setting, we see that altering the description of RLHF to avoid mentioning goals doesn’t substantially alter the rate of lying.

Interestingly, in the animal welfare setting, we find lying increases when cutting discussion of the AI company having a corporate partner (Jones Foods) trying to suppress animal welfare concerns (lying

¹¹³Except for the one which asks the AI for views about Anthropic.